INCH-POUND

MIL-T-47325(MI)
13 October 1989

MILITARY SPECIFICATION

# TAPE, PRESSURE-SENSITIVE ADHESIVE, ELECTRICAL INSULATION

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of pressure-sensitive, adhesive-coated, polyimide film tape used in electrical insulation applications.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5970

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SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-T-680 | Tape, Pressure-Sensitive Adhesive: Packaging and Packing of |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-130 | Identification Marking of U.S. Military Property |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

| | | |
|---|---|---|
| ASTM D 257 | - | Standard Methods of Test for DC Resistance or Conductance of Insulating Materials |
| ASTM D 1000 | - | Standard Methods of Testing Pressure-Sensitive Adhesive Coated Tapes Used for Electrical Insulation |

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103. Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Materials and construction.

3.2.1 Backing. The tape backing shall be amber-colored transparent polyimide film (see 4.5.1).

3.2.2 Adhesive. The tape shall be coated on one side with a pressure-sensitive silicone adhesive requiring no moisture, heat, or other preparation prior to or after application to clean dry surfaces (see 4.5.1).

3.3 Physical properties.

3.3.1 Thickness. The total thickness of the unstretched tape shall be 0.0030 (±0.0005) inches (see 4.5.2).

3.3.2 Width. The width of the tape shall be in accordance with table I as specified in the contract order (see 6.2).

TABLE I. Width codes.

| Width in inches | Tolerance Plus or minus | Letter code |
|---|---|---|
| .500 | .0075 | A |
| .750 | .0075 | B |
| 1.00 | .0156 | C |
| 1.25 | .0156 | D |
| 1.50 | .0156 | E |
| 1.75 | .0156 | F |
| 2.00 | .0156 | G |

3.3.3 Length. The tape shall be furnished in 36-yard (-0, +0.250) lengths (see 4.5.3).

3.3.4 Tensile strength. Tensile strength before stretching shall be 24 pounds per inch width or greater (see 4.5.4).

3.3.5 Elongation. Elongation at break shall be not less than 50 percent (see 4.5.5).

3.3.6 Adhesion to steel. Adhesion to steel shall equal or exceed 18 ounces per inch width when tested as specified (see 4.5.6).



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3.4 Electrical properties.

3.4.1 Insulation resistance. The insulation resistance of the tape shall be $1 \times 10^6$ megohms, minimum (see 4.5.7).

3.4.2 Dielectric breakdown. Dielectric breakdown of the tape shall occur at 7000 volts, minimum, when tested in accordance with ASTM D 1000 (see 4.5.8).

3.5 Indirect electrolytic corrosion. The tape shall have an indirect electrolytic corrosion as indicated by electrical conductance of not more than 1.0 pico ohms per inch width (see 4.5.9).

3.6 Flammability. Burning tape shall self-extinguish within 2 inches in length from the point of flame exposure when tested in accordance with 4.5.10 for flammability.

3.7 Rolls and cores.

3.7.1 Rolls. The tape shall be furnished in evenly and uniformly wound rolls, adhesion side in, on cores made of paper or nonfibrous plastic. Each roll shall consist of a single length of the tape, except that any single roll may contain not more than one splice. The splice shall be made so that the spliced ends will not separate when the roll is unwound for inspection or during application (see 4.5.1).

3.7.2 Cores. The cores shall have sufficient rigidity to prevent distortion of the rolls under normal conditions of transportation, storage, and use, and shall be sufficiently light in color to support marking requirements of 3.8. The inside diameter of the core shall be 3 inches (-0, +.0625) (see 4.5.1).

3.8 Marking. Marking for identification shall be in accordance with MIL-STD-130 and shall appear on the inner diameter of the core. Marking shall include product nomenclature, part or identifying number (PIN), title and date of this specification, manufacturer's name or trademark, and date of manufacture (see 4.5.1).

3.9 Workmanship. The tape shall be clean and free from foreign matter and defects that may impair its serviceability or appearance. The adhesive shall be a smooth uniform coating covering the entire area of one side of the backing. The edges of the tape shall be straight, true, and unbroken (see 4.5.1).



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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (4.4).

4.3 First article inspection. When specified in the contract or purchase order (see 6.2), a first article sample of two rolls of tape shall be provided. The sample shall be fabricated using the same processes, materials, and facilities as are proposed for production. The first article sample inspection shall consist of all the examinations and tests specified herein. Should a sample fail, it shall be rejected. No material shall be offered for acceptance until approval of a first article sample has been obtained from the procuring activity.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests shown in table II.



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TABLE II. Quality conformance inspection.

| Inspection | Requirement paragraph | Test paragraph |
|---|---|---|
| Backing | 3.2.1 | 4.5.1 |
| Adhesion | 3.2.2 | 4.5.1 |
| Thickness | 3.3.1 | 4.5.2 |
| Width | 3.3.2 | 4.5.3 |
| Length | 3.3.3 | 4.5.3 |
| Tensile strength | 3.3.4 | 4.5.4 |
| Elongation | 3.3.5 | 4.5.5 |
| Adhesion to steel | 3.3.6 | 4.5.6 |
| Insulation resistance | 3.4.1 | 4.5.7 |
| Dielectric breakdown | 3.4.2 | 4.5.8 |
| Indirect electrolytic corrosion | 3.5 | 4.5.9 |
| Flammability | 3.6 | 4.5.10 |
| Rolls | 3.7.1 | 4.5.1 |
| Cores | 3.7.2 | 4.5.1 |
| Marking | 3.8 | 4.5.1 |
| Workmanship | 3.9 | 4.5.1 |

4.4.1 Lot size. Lot size shall consist of all the tape manufactured in one production batch run, from one process, forming part of one contract or order, and submitted for acceptance at one time and place.

4.4.2 Sampling. A random sample of two rolls of tape shall be selected at the beginning and at the completion of the batch run. Each sample shall be subjected to quality conformance inspection. Failure of either sample shall be cause for rejection of the lot.

4.5 Methods of inspection.

4.5.1 Visual inspection. A visual inspection shall verify compliance with 3.2.1, 3.2.2, 3.7.1, 3.7.2 (except that core diameter shall be measured with a device capable of measuring in increments of 0.0625 inch), 3.8, and 3.9.

4.5.2 Thickness. The tape total thickness shall be determined in accordance with ASTM D 1000 to show compliance with 3.3.1.

4.5.3 Width and length. The tape width and length shall be measured in accordance with ASTM D 1000 for conformance to 3.3.2 and 3.3.3.



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4.5.4 Tensile strength. The tape tensile strength shall be determined in accordance with ASTM D 1000 to show compliance with 3.3.4.

4.5.5 Elongation. The tape elongation at break shall be determined in accordance with ASTM D 1000 to show compliance with 3.3.5.

4.5.6 Adhesion to steel. The tape adhesion to steel shall be determined in accordance with ASTM D 1000, method A, to show compliance with 3.3.6.

4.5.7 Insulation resistance. The insulation resistance shall be determined in accordance with ASTM D 257, except conditioning shall be in accordance with ASTM D 1000 to show compliance with 3.4.1.

4.5.8 Dielectric breakdown. The dielectric breakdown shall be determined in accordance with ASTM D 1000 to show compliance with 3.4.2.

4.5.9 Indirect electrolytic corrosion. The test for indirect electrolytic corrosion shall be conducted in accordance with ASTM D 1000 to show compliance with 3.5.

4.5.10 Flammability. The test for flammability shall be conducted in accordance with ASTM D 1000 except the tape shall be wound on the rod for greater than six inches in length to show compliance with 3.6.

4.6 Rejection. Individual lots or units, as applicable, which do not meet the requirements of this specification shall be rejected.

5. PACKAGING

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with the requirements of PPP-T-680, level A, B, or C as specified in the procurement documents (see 6.2).

5.2 Marking. Containers of the tape shall be marked in accordance with the requirements of PPP-T-680 for the type of shipment specified in the procurement documents (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The tape is intended for use in electrical insulation applications, especially high performance applications requiring high-temperature retention of physical and electrical properties, mechanical strength, and flame retardance.



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grime, grease, and abraded or damaged surfaces

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. The part or identifying number

d. Whether a first article sample is required (see 3.1 and 4.3)

e. Applicable level(s) of preservation, packaging, and packing (see 5.1)

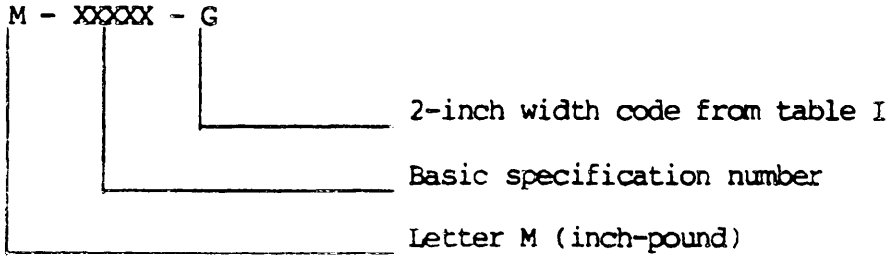
f. Marking and shipping instructions (see 5.2).

6.3 Storage. The tape is to be stored in its original container, preferably in a cool location. It is not to be stored in close proximity to steam pipes, radiators, or other sources of heat. For best results, store tape at 21° Celsius (C) (70° Fahrenheit (F)) and 40 to 50 percent relative humidity.

6.4 First article. When a first article inspection is required, the items should be a first article sample consisting of two rolls of tape. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Part or identifying number (PIN). The PIN shall consist of the letter M, the basic number of this specification, and dash numbers denoting width (refer to table I).

Example:






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6.6 Metrication. Whenever inch-pound dimensions are used in this document, metric equivalents in accordance with FED-STD-376 shall be acceptable.

6.7 Subject term (key word) listing.

Attaching materials
Binding materials
Filament
Ribbon
Securing materials
Strapping materials

Custodian:
Army - MI

Preparing Activity:
Army - MI

Project No. 5970-A669



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*NOTE:* This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-T-47325 | |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)* _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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